DM DW QUESTION BANK

UNIT I

1) Explain the architecture of a typical data mining system with a neat diagram. Describe the role of each component in the architecture.

2) Compare different classifications of data mining systems based on data type, application area, and mining methodology. Analyze their advantages and limitations.

3) Explain with a neat diagram how a Star Schema organizes fact and dimension tables in a data warehouse.

4) Compare Star Schema and Fact Constellation Schema in terms of complexity, flexibility, and storage requirements.

5) Examine how schema choice (Star vs Snowflake) affects OLAP operations such as roll-up, drill-down, and slice-dice.

6) Compare ETL (Extract–Transform–Load) tools based on their capabilities in extraction, cleansing, and transformation.

7) Explain the role of Data Extraction, Clean-up, and Transformation tools in preparing data for OLAP operations. Illustrate with a suitable diagram.

UNIT II

1) Explain the architecture of a typical Data Mining system. Describe the function of each major component involved in the process.

2) Analyze different data mining functionalities and classify them into descriptive and predictive categories.

3) Compare data cleaning and data reduction techniques.

4) Explain the various data preprocessing steps involved in preparing raw data for mining.

5) Illustrate the process of data integration with an example.

6) Explain the role of ETL (Extract, Transform, Load) in data preprocessing.

7) Explain data cleaning, integration, and transformation.

Unit III

1) Define the following terms: Frequent Itemset, Support, Confidence, Classification, Prediction, Lazy Learner.

2) Explain the Apriori algorithm for frequent itemset mining. Describe how support and confidence thresholds influence association rules.

3) Explain the concept of Lazy Learning with reference to K-Nearest Neighbor (KNN). Why is it called "lazy"?

4) Apply the Apriori algorithm on a given small dataset (provide your own example) to find frequent itemsets and generate association rules.

5) Compare Apriori and FP-Growth algorithms for frequent itemset mining. Analyze their strengths and weaknesses in handling large datasets.

6) Analyze how Support Vector Machines differ from other classification techniques in terms of decision boundaries, computational complexity, and data requirements.

7) Given the following transaction dataset, compute all frequent itemsets using the Apriori algorithm with minimum support = 50%.

TID Items
T1 A, B, C
T2 A, C
T3 A, B
T4 B, C
T5 A, B, C

(Show L1, L2, L3 itemsets.)

8) A retail store has the following item frequencies:

A: 20, B: 18, C: 15, AB: 12, AC: 10, BC: 9, ABC: 7
If min-support = 40%, identify which itemsets are frequent assuming total transactions = 25.

9) Using the following dataset, construct a decision tree using Information Gain (Entropy).

Age	Income	Student	Buys Laptop
Young	High	No	No
Young	High	Yes	Yes
Middle	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Medium	No	No

10) For the following points, classify the test point (3,3) using KNN with K=3. Compute Euclidean distance and majority class.

Point	Class
(2, 4)	A
(4, 4)	A
(3, 2)	B
(6, 5)	B

Point	Class
(1, 3)	A

11) Consider the following dataset of students with attributes *(Study Hours, Attendance, Result)*:

Study Hours	Attendance	Result
5	85	Pass
3	70	Pass
2	60	Fail
6	90	Pass
1	50	Fail

Using **K = 3**, classify a new student with **Study Hours = 4 and Attendance = 80 using the K-Nearest Neighbor (KNN)** algorithm.

Unit IV

1) List and describe the major categories of clustering methods: Partitioning, Hierarchical, Density-based, Grid-based, and Model-based approaches.

2) Explain how each major clustering method category differs in terms of algorithmic approach and outcome.

3) Define partitioning methods and describe the basic idea behind algorithms such as K-means and K-medoids.

4) Explain how the K-means algorithm works with a step-by-step example. Describe its limitations.

5) Define hierarchical clustering. Distinguish between agglomerative and divisive approaches.

6) Explain how DBSCAN identifies core points, border points, and noise. Provide an illustrative example.

7) A spatial database contains the following points on a 2D grid: (1,2), (2,2), (2,3), (8,7), (8,8), (25,80) Apply DBSCAN with parameters Eps = 2 and MinPts = 2. Identify core points, border points, and noise points.

Unit V

1) Define spatial data mining and list its major components (spatial objects, spatial relations, spatial attributes).

2) Analyze the challenges associated with mining multimedia data, including high dimensionality, semantic gap, and heterogeneous formats.

3) Explain the roles of tokenization, stemming, stop-word removal, and TF–IDF in text mining.

4) Explain how web usage mining works using server logs, session data, and clickstream patterns

5) How can e-commerce companies integrate Text Mining and Web Mining to build an effective customer sentiment analysis system? Present a detailed workflow describing the integration.

6) Write short notes on the following:
   i) Mining heterogeneous data sources
   ii) Mining dynamic web and multimedia streams

7)) How does OLAP support analysis of spatial, multimedia, and web data?

8) Create a framework that shows how spatial mining and multimedia mining can work together in a smart-city surveillance system. Explain each component, how data moves through the system, and the results expected from such analysis."